

Evidence-Traced Meeting Assistant

Technical Report

ML Bootcamp Project

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Abstract

We present a meeting assistant that converts a recorded meeting into a raw transcript, a terminology-refined transcript and a structured meeting record (summary, minutes, decisions and action items). Instead of chaining a speech recogniser with free-form language-model prompts, the system treats every language model as a *proposer* whose output must be supported by evidence before it is kept. Transcript corrections are restricted to minimal span edits that must pass deterministic checks, a second language model acting as a veto-only supervisor, and an acoustic test in which the speech recogniser scores the audio with and without the edit. Meeting documentation is produced from cited dialogue events (proposals, agreements, commitments) whose lifecycle is tracked across the whole meeting by code; decisions, owners, deadlines and numbers are accepted only if they occur in the cited transcript lines. Every item in the final record carries segment identifiers, timestamps and verbatim quotes, and can be played back from the audio. The system uses three distinct language models for refinement, supervision and documentation, tested on a 6 GB laptop GPU plus a hosted inference API, and is delivered as a staged web application.

1 Introduction

Meeting records are only useful if they are faithful: a summary that presents a suggestion as a decision, or assigns a task to someone who never accepted it, is worse than no summary at all. The project brief requires an English meeting recording to be processed end to end into (i) a raw transcript, (ii) a refined transcript in which domain-specific recognition errors are corrected while names, numbers, negations and commitments are preserved, and (iii) minutes, key decisions and action items, where an owner or deadline is reported only when it was stated. The two language-model roles (refinement and documentation) must be distinct stages.

The straightforward solution is *Whisper* $\rightarrow$ *prompt* $\rightarrow$ *prompt* $\rightarrow$ *JSON*. It is easy to build but every step trusts the previous one: *Whisper* can produce text during silence; a “fix the transcript” prompt can silently change numbers, negations or names; and a “write the minutes” prompt can invent owners, normalise deadlines or report proposals as decisions, with no way for a reader to verify any claim.

Our design is built around one principle: **every word in the output must be traceable back to the audio**. Concretely:

- language models return *structured claims with citations*, never final text;
- deterministic code decides which claims survive, using evidence from the transcript (verbatim quotes, word confidences) and from the audio (acoustic likelihoods, speaker embeddings);
- uncertainty is made visible (**unspecified** owners, “proposals not adopted”, a warnings log) instead of being hidden behind fluent text.

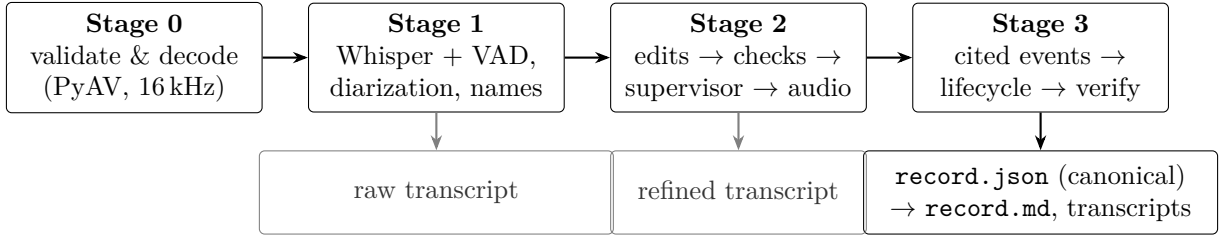


Figure 1: Pipeline. Each stage can be run separately from the web interface and shows its output when finished.

2 System overview

Table 1 lists the models. Local models are loaded one at a time and released before the next one, which keeps peak GPU memory within 6 GB (tested on an NVIDIA RTX 3050 laptop GPU).

Role	Model	Purpose
Speech-to-text	faster-whisper large-v3-turbo	Transcription with word timestamps and confidences
Diarization	pyannote/speaker-diarization-3.1	Who spoke when
Voice re-check	wespeaker-voxceleb-resnet34-LM	Sentence-level speaker verification
LLM-1 (refiner)	openai/gpt-oss-20b	Proposes glossary terms and minimal edits
LLM-2 (supervisor)	qwen/qwen3.8-27b	Approves or rejects every edit (veto only)
Acoustic verifier	openai/whisper-large-v3-turbo	Teacher-forced scoring of edits
LLM-3 (documenter)	openai/gpt-oss-120b	Speech acts, decision/task events, minutes, speaker evidence
NLI (optional)	nli-deberta-v3-small	Flags weakly supported items

Table 1: Models and their roles. The three language models are served by the Groq API.

3 Stage 0 and Stage 1: input and transcription

Input validation. Uploaded files are probed with PyAV before any model runs: supported container, non-empty, readable, containing an audio stream, at least 0.5s long and not silent. Failures produce a specific user-facing message. Accepted audio is decoded to 16 kHz mono.

Speech recognition. We run faster-whisper with voice-activity detection (Silero VAD) before decoding, so no text is generated over silence, and with conditioning on previous text disabled, so a hallucination cannot propagate into the next 30-second window. Word-level timestamps and probabilities are kept: they define which words are uncertain (Stage 2) and which audio to play back (interface). User-supplied agenda terms and attendee names are passed to Whisper as hotwords. A small normalisation re-joins decimals and abbreviations that Whisper emits as separate tokens (e.g. “12 .50”).

Speaker diarization and voice re-check. Words are assigned to speakers by temporal overlap with pyannote’s exclusive (one speaker at a time) diarization, and single-word speaker flips are smoothed. Diarization can fold a short interjection into the long turn around it, so we add a *sentence-level voice re-check*. For each speaker k we compute a centroid $\mathbf{c}_k$ of L2-normalised embeddings over 3-second chunks of that speaker’s own long turns. Each sentence u (split at sentence punctuation, pauses > 0.8 s and speaker changes, duration ≥ 0.6 s) is embedded as $\mathbf{e}_u$ and moved from its current speaker c to speaker $b = \arg \max_k \mathbf{e}_u^\top \mathbf{c}_k$ only if

$$\mathbf{e}_u^\top \mathbf{c}_b \geq 0.45 \quad \text{and} \quad \mathbf{e}_u^\top \mathbf{c}_b - \mathbf{e}_u^\top \mathbf{c}_c \geq 0.20. \quad (1)$$

Segments are then split at speaker changes.

Speaker naming from evidence. Speakers keep anonymous labels unless the meeting itself names them. LLM-3 reports four kinds of evidence with a quote and a segment identifier: a self-introduction, being addressed by name and answering in the next turns, a self-stated role, and a role stated about a named person. Unambiguous “my name is ...” statements are additionally detected by pattern. Every claim is validated by code:

- the name (or role) must appear verbatim, up to fuzzy matching, in the cited segment. If the model cited the wrong segment, the claim is *re-anchored* to the nearest segment that does contain the quote;
- a self-introduction must be spoken by the claimed speaker and the name must follow an introducing phrase (a mere mention is rejected); an “addressed” claim requires the named label to answer within two turns;
- a role must be a short noun phrase, not an activity.

Names are chosen per label by a weighted vote (self-introduction = 3, addressed = 1). A first name and a full name containing it count as one person; a name that two labels claim equally is left unassigned. A word spelled out letter by letter (“X-Y-Z”) overrides how the name was transcribed if the two sound alike under Double Metaphone, because a spelling is the strongest evidence of a name a recording can contain.

4 Stage 2: evidence-gated refinement

LLM-based ASR correction is effective but, used as free rewriting, cannot guarantee that meaning is preserved. Our refiner is constrained to *minimal span edits* of the form ⟨original words, replacement, type, reason, confidence⟩; it never returns rewritten text.

Glossary and candidate spans. The glossary combines user terms, attendee names, words spelled out in the meeting (treated as verified spellings) and terms proposed by LLM-1. A proposed term is kept only if it occurs in the transcript or its claimed misheard form does. Candidate spans are the union of (a) runs of words with Whisper probability below 0.5 and (b) word n -grams ($n \leq 3$) whose Double Metaphone code is within an optimal-string-alignment similarity of 0.8 of a glossary term. Candidates guide the refiner and determine how much evidence an edit needs; they do not decide on their own whether a span may be edited.

Deterministic checks. An edit is rejected if its original words are not found in the segment (hallucination), if it only deletes words, if its replacement is much longer than the original (a rewrite), or if it merely capitalises an ordinary word. An edit labelled “formatting” must keep the same spoken letters; otherwise it is treated as acoustic. Edits that change a number (spelled-out numbers included), a negation, a modal or commitment word, or a person’s name are marked *protected*.

Supervisor. Every remaining edit is reviewed by LLM-2, a different model, which sees the transcript before and after the edit and must confirm correct casing, grammar no worse than before, and unchanged meaning. Corrections towards a verified spelling count as meaning-preserving. The supervisor can only veto; a missing review counts as a rejection, so an unreviewed edit never reaches the transcript. This follows the LLM-as-a-judge paradigm, restricted to approval.

Acoustic verification. The Whisper checkpoint used for recognition (via Hugging Face Transformers) scores the full segment text y and the edited text y' under teacher forcing given the

segment audio x , with the same glossary prompt for both. Because both hypotheses explain the same audio, total log-likelihoods are compared without length normalisation:

$$\Delta = \log p(y' | x) - \log p(y | x). \quad (2)$$

An ordinary edit with mean word confidence $\bar{p}$ over the replaced span is accepted if

$$\Delta \geq -(\tau_c(1 - \bar{p}) + \tau_g g), \quad \tau_c = 6.0, \tau_g = 3.0, \quad (3)$$

where $g = 1$ if the replacement is a glossary term that sounds like the original. The tolerance compensates for Whisper’s language-model prior, which penalises rare domain terms. A protected edit must be *preferred* by the audio: $\Delta \geq m = 0.5$. Formatting-only edits skip this test. When the acoustic model is unavailable, a conservative fallback applies and protected edits are always rejected.

Application and logging. Accepted edits are applied right to left within each segment, after resolving overlapping edits. Every proposal is logged with its evidence, confidence, Δ , protected-word hits, supervisor verdict and final reason, and is shown in the interface as a diff and a refinement log.

5 Stage 3: documentation from cited events

Decisions and action items in multi-party dialogue are best modelled as structures of dialogue acts rather than as isolated sentences. LLM-3 therefore processes the refined transcript in chunks of about 600 words and emits, for each chunk:

- speech acts per segment: proposal, agreement, objection, decision, commitment, assignment, question;
- *events* with a segment identifier and a verbatim quote: new announced decisions, new proposals, proposal status updates, new tasks and task updates.

Lifecycle tracked by code. Each proposal moves through *proposed* $\rightarrow$ *accepted* / *rejected* / *deferred* / *unresolved*. Open proposals and tasks are carried into the next chunk, so a proposal can be resolved much later in the meeting. An acceptance is honoured only if its segment is tagged as an agreement, decision or commitment, and not if it comes from the proposer alone. Decisions are proposals that reach *accepted*, plus facts presented as settled (targets, limits, requirements), which are marked *announced*. Everything else is reported under “proposals not adopted”, never as a decision.

Tasks. Only follow-up work counts; activities performed during the meeting are discarded. Tasks are self-commitments, assignments, open tasks or requests. A request that is never accepted is listed separately as unconfirmed, and a task restated a few lines later is merged.

Robust grounding. Every event’s quote must be found in the cited segment or its neighbours. If the model cited the wrong segment, the event is re-anchored to the chunk segment that contains the quote rather than dropped. A *consistency pass* sends one follow-up request when a segment was tagged as a proposal, decision, assignment or commitment but no event cites it.

Summary and minutes. These are written last, from the transcript (or from map-reduced notes for long meetings) together with the *verified* lists, so that the prose cannot contradict them. Every minutes point cites segment identifiers.

6 Stage 4: verification

- **Owners.** An owner is either a name or role spoken in the cited lines (copied verbatim from the transcript, not from the model’s wording) or the speaker of a first-person commitment. That speaker is shown by name only if they introduced themselves. A name inferred from context is kept as an annotation, never as the owner. If diarization split the commitment sentence between two speakers, the owner is left **unspecified**.
- **Deadlines** are copied verbatim (“by next week”) and never converted to dates.
- **Numbers.** Every number in a model-written decision or task description must occur in the cited lines (spelled-out numbers are canonicalised), otherwise the item is flagged.
- **NLI flags** (optional). A DeBERTa-v3 cross-encoder trained on SNLI/MultiNLI flags items whose wording is not entailed by their cited lines, following NLI-based faithfulness checking. Flags never delete items.

7 Implementation

Language-model client. All three models are called through one client that:

- requests JSON-only replies and validates them against Pydantic schemas, with repair retries;
- tolerates common quirks such as nested objects returned as JSON strings;
- paces requests to the tokens-per-minute limit and caps output budgets so that a request never exceeds the free-tier limit;
- lowers the reasoning effort automatically when a reasoning model spends its entire output budget thinking;
- caches only replies that validated.

General prompts. The prompts define every category by its function in a conversation and contain no names, numbers or phrases from any particular meeting, so behaviour does not depend on a recording having been seen during development.

Outputs and interface. The canonical record is a single JSON document containing the summary, minutes, decisions, action items, unconfirmed requests, proposals not adopted, speakers, both transcripts, every edit verdict and all warnings. The Markdown report is rendered from it, so both formats always agree; empty decision or task lists are valid outputs. The Streamlit interface runs the stages one by one with live status, shows transcripts as one paragraph per speaker turn, and lets the user play the audio behind every decision and task.

Testing. Twenty-one offline tests (no GPU, models or API key; language models are replaced by fixed replies) cover the guards, the edit gates, the supervisor veto, the lifecycle and owner rules, speaker naming (roles, name variants, spelled names, wrong segment identifiers), the number check and reply recovery.

8 Limitations and future work

- **Precision over recall.** The design deliberately omits items without sufficient evidence. Tentatively phrased decisions can be reported as unresolved proposals. Measuring this trade-off on held-out meetings and refining the general definition of a settled decision is the main next step.

- **Dependence on diarization.** Speaker names and self-commitment owners depend on diarization, which degrades with overlapping speech and distant microphones.
- **Ambiguous spoken forms.** Some spoken numbers (e.g. “twelve fifty”) and unspelled names cannot be resolved from audio alone; they are corrected only when other evidence exists.
- **Thresholds** for the acoustic test and the voice re-check were set on few examples and should be tuned on more recordings.
- **Cost.** The pipeline makes more language-model calls than a two-prompt baseline and is subject to free-tier rate limits.

9 Conclusion

Rather than asking language models to be right, the system assumes they will sometimes be wrong and makes their errors either impossible to commit (edits must exist verbatim, protected words need audio support, a supervisor can veto) or visible to the reader (**unspecified** fields, proposals not adopted, flags and a warnings log). Every decision and action item in the final record carries the quotes, segment identifiers and timestamps needed to check it against the recording.